Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In the expression 5x + 12, what is the constant term? Student's answer: 5x

Identify the part of the expression with no variable.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In the expression 7 - 2t, what is the coefficient of t? Student's answer: 2

Include the sign with the coefficient.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

n costs 25 + 10g dollars, where g is the number of GB used. What does 25 represent? Student's answer: the

Decide which part does not change with usage.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ength l and width w. The expression 2(l + w) gives the perimeter. What does l + w represent? Student's answ

Think about how the perimeter is written.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ation is P = 800 × 0.9^t, where t is the number of years. What does 0.9 represent? Student's answer: 90% gro

Decide whether the population grows or shrinks.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What are the terms of $4x^2 - 3x + 9$? Student's answer: $4x^2$, $3x$, 9

Keep the sign with each term.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $x^2 - 9$. Student writes: $(x - 3)(x - 3)$

Look for two numbers that multiply to –9 and add to 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $x^2 + 5x + 6$. Student writes: $(x + 1)(x + 6)$

Find two numbers that multiply to 6 and add to 5.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $4x^2 - 25$ as a difference of squares. Student writes: $(4x - 5)(x + 5)$

Write each term as a square.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $3x^2 + 6x$ completely. Student writes: $3(x^2 + 2x)$

Look for every common factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Use the difference of squares to find 51 × 49 without a calculator. Student's answer: 2,500

Write 51 and 49 as 50 + 1 and 50 − 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor $x^2 - 16$. Student writes: $(x - 4)^2$

Check by multiplying.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(x + 3)(x + 5)$. Student writes: $x^2 + 15$

Use FOIL and combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add $(2x - 1) + (3x + 4)$. Student writes: $5x + 5$

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(x + 2)(x - 2)$. Student writes: $x^2 + 4$

Multiply each pair of terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract $(3x^2 + x) - (x^2 - 4x)$. Student writes: $2x^2 - 3x$

Distribute the negative sign to both terms of the second polynomial.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Expand $(x - 4)^2$. Student writes: $x^2 + 16$

Write the square as a product of two binomials.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(2x + 1)(x - 3)$. Student writes: $2x^2 - 6x - 3$

Multiply each pair of terms, then combine.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

A.CED.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an equation: a number increased by 7 is 19. Student writes: $x - 7 = 19$

Translate the words into operations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

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Find the Error

A.CED.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A taxi charges $3 plus $2 per mile. The total fare is $15. Write an equation. Student writes: $2(m + 3) = 15$

Decide which part is multiplied by the miles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

bakery sells 12 cookies per day. Write an inequality for at least 120 cookies in d days. Student writes: $12d \leq 12$

Think about the words at least.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an equation: twice a number minus 4 is 18. Student writes: $2x + 4 = 18$

Translate each phrase in order.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

A.CED.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ctangle has a length 3 more than its width w. The perimeter is 30. Write an equation. Student writes: 2w + 3

Use the perimeter formula with both lengths and widths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pens cost p dollars each, plus a $5 bag. The total is at most $25. Write an inequality. Student writes: 4p + 5

Think about the words at most.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the point (2, 5) on the graph of $y = 2x + 1$? Student says: No

Substitute 2 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line passes through (0, 3) and (1, 5). Write its equation. Student writes: $y = 3x + 2$

Find the slope, then the y-intercept.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the y-intercept of $y = -x + 4$? Student's answer: -1

Find the value of y when x = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A car travels at 55 miles per hour. Let x be hours and y be miles. Write an equation. Student writes: y = x/55

Decide whether miles depend on hours.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the point (3, -2) on the graph of $y = -x + 1$? Student says: No

Substitute 3 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which equation is parallel to y = 3x - 2? Student writes: y = -3x + 5

Parallel lines have the same slope.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

st $5. The total revenue is $200. Let a be the number of adult tickets and s the number of student tickets. W

Multiply each count by its price.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e needs at least $500 by selling x items at $12 and y items at $8. Write an inequality. Student writes: 12x + 8y

Think about the words at least.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A class has 30 students. There are 6 more girls g than boys b. Write the equations. Student writes: g = b - 6

Write the relationship in words first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

osts $3 per meter, and you can spend no more than $240. Write an inequality for m meters. Student writes:

Think about the words no more than.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$2 and pencils for $1. A student spends at most $20 and buys at least 4 pens. Let p be pens and q be pencil

Write each condition as an inequality.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

rden has an area of at most 60 square feet with length l and width w. Write an inequality. Student writes: lw

Think about the words at most.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve A = lw for w. Student writes: w = Al

Undo the multiplication by l.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $P = 2l + 2w$ for w. Student writes: $w = (P + 2l)/2$

Subtract 2l first, then divide by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve d = rt for t. Student writes: t = dr

Undo the multiplication by r.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $V = (1/3)Bh$ for h. Student writes: $h = V/(3B)$

Multiply by 3, then divide by B.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $C = (5/9)(F - 32)$ for F. Student writes: $F = (9/5)C - 32$

Multiply by 9/5, then undo the subtraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $y = mx + b$ for x. Student writes: $x = y - b/m$

Subtract b, then divide by m.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3x + 2 = 8$. A student's first step is: $3x = 10$. Is this step correct? Student says: Yes

Check the first step.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lain why adding 2 to both sides of x - 2 = 5 keeps the solution the same. Student says: It changes the soluti

Think about what the equal sign means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $\sqrt{x} = -2$. Student's answer: $x = 4$

Square both sides and check the result.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2(x - 1) = 8$. Student's answer: $x = 4.5$

Distribute, then solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $5x = 3x + 6$. Student's answer: $x = 6$

Subtract 3x from both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is multiplying both sides of an equation by 0 a valid step when solving? Student says: Yes

Think about whether the solution is kept.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3x - 4 < 11$. Student writes: $x > 5$

Add 4, then divide by 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $-2x + 3 \geq 9$. Student writes: $x \geq -3$

Subtract 3, then divide by −2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2(x + 4) > 2x + 8$. Student says: All real numbers are solutions.

Distribute, then simplify both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $4x - 2 = 4x + 6$. Student's answer: $x = 2$

Subtract 4x from both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x/4 - 1 \leq 2$. Student writes: $x \leq 3$

Add 1, then multiply by 4.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $-(x - 5) = 2x$. Student's answer: $x = -5/3$

Distribute the negative sign, then collect the x terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 = 49$. Student's answer: $x = 7$

Take the square root of both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $(x - 3)^2 = 16$. Student's answer: $x = 7$

Take the square root of both sides, including the negative root.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + 6x + 9 = 0$. Student's answer: $x = 3$

Factor the trinomial.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 - 5x + 6 = 0$. Student's answer: $x = 1$ or $x = 6$

Find two numbers that multiply to 6 and add to -5.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 + 4x = 5$ by completing the square. Student's answer: $x = -2 \pm \sqrt{5}$

Add the square of half the coefficient to both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 - 2x - 8 = 0$ with the quadratic formula. Student's answer: $x = 2$ or $x = -4$

Use $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + y = 10$ and $2x - y = 8$. Student's answer: $(4, 6)$

Add the equations to eliminate y.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $2x + 3y = 12$ and $x - y = 1$. Student's answer: $(2, 3)$

Solve one equation for a variable and substitute.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + y = 5$ and $x + y = 7$. Student's answer: $(1, 4)$

Compare the two equations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $3x - y = 4$ and $6x - 2y = 8$. Student's answer: $(2, 2)$

Multiply the first equation by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is (1, 2) a solution of the system $2x + y = 4$ and $x - y = -1$? Student says: No

Substitute into both equations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The system $4x + 2y = 10$ and $2x + y = 5$ is solved. Student says there is exactly one solution, $(2, 1)$.

Multiply the second equation by 2 and compare.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = 2x + 1$ and $y = -x + 7$. Student's answer: (1, 3)

Set the two expressions equal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = 3x - 2$ and $y = 3x + 4$. Student's answer: (3, 7)

Compare the slopes and intercepts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $y = 2x$ and $x + y = 9$ by substitution. Student's answer: (6, 3)

Substitute 2x for y.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $y = -2x + 8$ and $y = x - 1$. Student's answer: (2, 3)

Set the two expressions equal and solve for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many solutions does the system $y = 4x + 1$ and $y = 4x - 3$ have? Student's answer: 1

Compare the slopes and intercepts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + 2y = 8$ and $x - y = 2$. Student's answer: $(2, 3)$

Subtract the second equation from the first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = 2x + 3$, what is $f(4)$? Student's answer: 8

Substitute 4 for x, then calculate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table lists x = 1, 2, 2 with y = 3, 4, 5. Is this a function? Student says: Yes

Check whether any input has two outputs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = x^2$, what is $f(-3)$? Student's answer: -9

Square the input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the domain of $f(x) = 1/(x - 2)$? Student's answer: all real numbers

Find any values that make the denominator zero.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the range of $y = x^2$ when x is any real number? Student's answer: all real numbers

Think about the smallest output the squares can give.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $g(x) = 5 - x$, what is $g(2)$? Student's answer: 7

Substitute 2 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sequence is defined by $a(n) = 3n + 1$. What is $a(4)$? Student's answer: 12

Substitute 4 for n.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A recursive sequence has $a(1) = 2$ and $a(n) = a(n - 1) + 3$. What is $a(3)$? Student's answer: 9

Find each term from the one before it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = x^2 - 1$, what is $f(3)$? Student's answer: 10

Substitute 3 and simplify.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table gives f(0) = 5, f(1) = 7, and f(2) = 9. What is f(0)? Student's answer: 7

Read the value from the table at x = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = 2$, what is $f(3)$? Student's answer: 6

Raise 2 to the power of 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sequence is 2, 6, 18, 54. What is the next term? Student's answer: 108

Find the rule that gets from one term to the next.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the y-intercept of $y = 3x - 6$? Student's answer: 6

Find y when x = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the x-intercept of y = 2x - 8? Student's answer: -8

Set y = 0 and solve for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the maximum value of $y = -x^2 + 4$? Student's answer: 4 is the minimum.

Decide whether the parabola opens up or down.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Over what interval is y = 2 increasing? Student's answer: it is decreasing everywhere

Look at the direction of the graph.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What are the zeros of $f(x) = x^2 - 9$? Student's answer: $x = 3$

Set f(x) = 0 and solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line passes through (0, 4) and (2, 0). What is its rate of change? Student's answer: 2

Use (change in y) ÷ (change in x).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s 100 gallons and drains 4 gallons per minute. Write f(t) for the amount after t minutes. Student writes: f(t)

Decide whether the amount goes up or down.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ccount starts with $20 and gets $5 each week. Write f(w) for the amount after w weeks. Student writes: f(w

Identify the starting amount and the weekly change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sequence is 3, 6, 12, 24. Write an explicit formula for the nth term. Student writes: $a(n) = 3n$

Check the ratio between terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $f(x) = x + 2$ and $g(x) = 3x$, what is $f(g(2))$? Student's answer: 12

Find g(2) first, then apply f.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

worker earns $15 per hour plus a fixed $30 bonus. Write the function for h hours. Student writes: f(h) = 30h +

Decide which part depends on hours.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The formula $F = (9/5)C + 32$ converts Celsius to Fahrenheit. What is F when $C = 20$? Student's answer: 52

Substitute and multiply first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has x = 0, 1, 2 and y = 2, 4, 8. Is it linear or exponential? Student's answer: linear

Check whether the outputs have a constant difference or a constant ratio.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = 3, 7, 11 for x = 0, 1, 2. Is it linear or exponential? Student's answer: exponential

Check the differences.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = 5, 10, 20 for x = 0, 1, 2. Is it linear or exponential? Student's answer: linear

Check the ratio between outputs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

100

Find the Error

F.LE.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A population increases by 10% each year. Is the model linear or exponential? Student's answer: linear

Decide whether the change is a fixed amount or a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

101

Find the Error

F.LE.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ccount starts with $500 and gains $20 each month. Is the model linear or exponential? Student's answer: e

Decide whether the change is a fixed amount or a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = 2, 6, 18 for x = 0, 1, 2. Is it linear or exponential? Student's answer: linear

Check the ratio and the difference.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write a linear function through (1, 5) and (3, 9). Student writes: $y = 5x + 1$

Find the slope, then the y-intercept.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A population of 100 grows by 50% each year. Write the exponential model. Student writes: y = 100(0.5)^t

Decide whether the growth is up or down.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

tial function passes through (0, 3) and doubles each time x increases by 1. Write the function. Student write

Start with the value at x = 0, then multiply by the growth factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A tank starts with 40 gallons and loses 4 gallons per year. Write a linear model. Student writes: $y = 40 + 4t$

Decide whether the amount increases or decreases.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A value of 500 decreases by 8% each year. Write the exponential model. Student writes: y = 500(1.08)^t

Decide whether the amount grows or decays.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line has slope -3 and passes through (2, 1). Write the equation. Student writes: $y = 3x + 7$

Use y = mx + b and the point to find b.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the median of the data 3, 5, 5, 7, 9. Student's answer: 6

Find the middle value of the ordered data.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

110

Find the Error

S.ID.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which display is best for categorical data, such as favorite colors? Student's answer: histogram

Think about what a histogram shows.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box plot has Q1 = 3 and Q3 = 9. What is the interquartile range? Student's answer: 12

Subtract Q1 from Q3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

112

Find the Error

S.ID.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A histogram has bars with frequencies 2, 5, and 3. How many data values are there? Student's answer: 8

Add all the frequencies.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a stem-and-leaf plot, 4 | 1 means what value? Student's answer: 4.1

Combine the stem and the leaf.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lata set has minimum 2, Q1 = 4, median = 6, Q3 = 8, and maximum 14. What is the range? Student's answer

Subtract the minimum from the maximum.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

andard deviation of 2. Group B has a mean of 55 and a standard deviation of 2. Which group has the highe

Compare the means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

andard deviation of 1 and Group B has a standard deviation of 5. Which group has more spread? Student's

Compare the standard deviations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

a set has an outlier of 100. Which center is less affected by the outlier: mean or median? Student's answer: r

Think about how an outlier changes a total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

value of 80 is in a data set with mean 70 and standard deviation 5. What is its z-score? Student's answer: 1

Use z = (value - mean) ÷ standard deviation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

eviation of 3. Group B has a mean of 14 and a standard deviation of 3. How many standard deviations is B's

Divide the difference in means by the standard deviation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

120

Find the Error

S.ID.2

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pproximately normal. About what percent of values lie within 2 standard deviations of the mean? Student'

Recall the empirical rule.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.